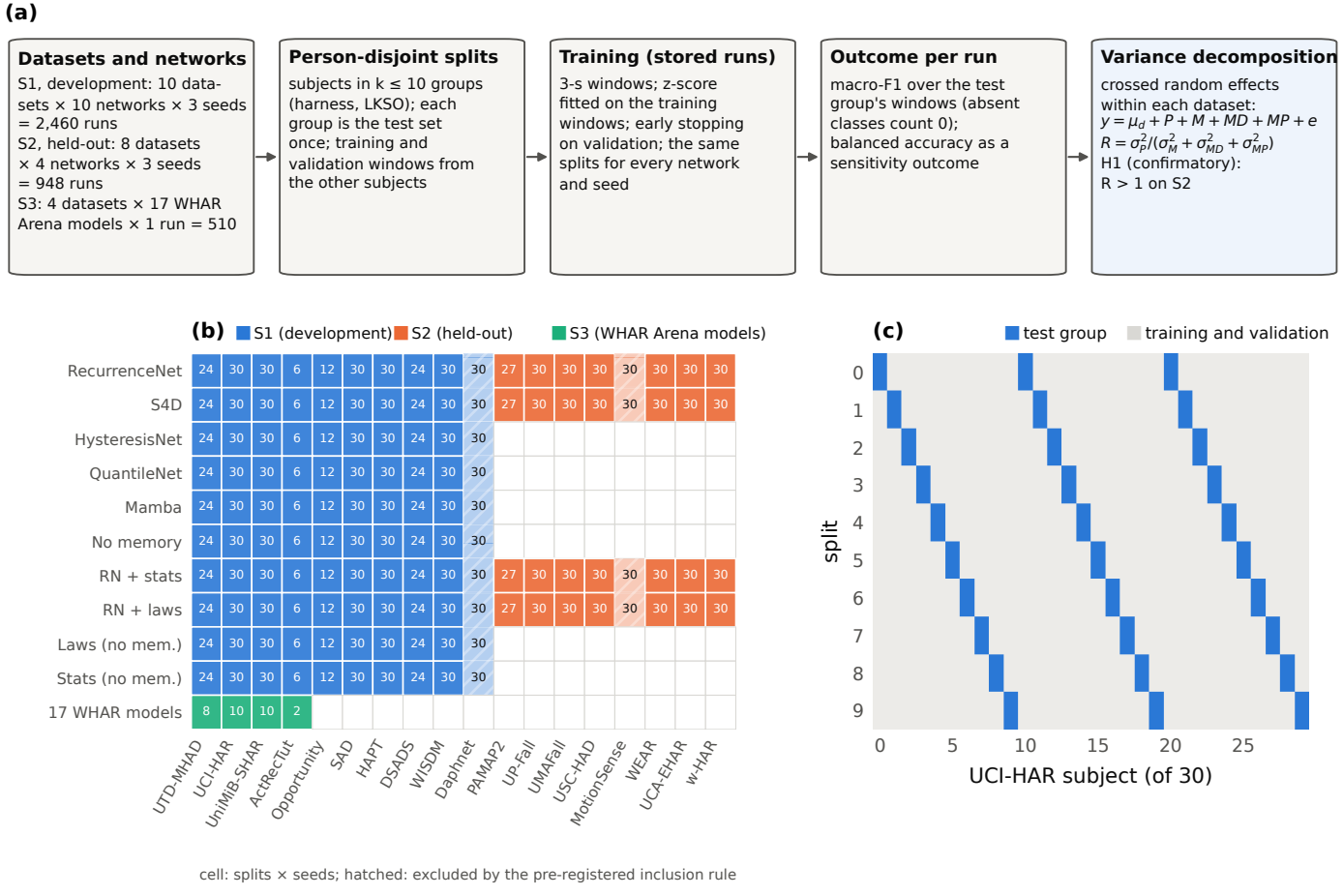
(a)
Datasets and networks
S1, development: 10 data-sets × 10 networks × 3 seeds = 2,460 runs
S2, held-out: 8 datasets × 4 networks × 3 seeds = 948 runs
S3: 4 datasets × 17 WHAR Arena models × 1 run = 510
Person-disjoint splits
subjects in k ≤ 10 groups (harness, LKSO); each group is the test set once; training and validation windows from the other subjects
Training (stored runs)
3-s windows; z-score fitted on the training windows; early stopping on validation; the same splits for every network and seed
Outcome per run
macro-F1 over the test group's windows (absent classes count 0); balanced accuracy as a sensitivity outcome
Variance decomposition
crossed random effects within each dataset:
$y = \mu_d + P + M + MD + MP + e$
$R = \sigma_P^2/(\sigma_M^2 + \sigma_{MD}^2 + \sigma_{MP}^2)$
H1 (confirmatory):
R > 1 on S2
(b)
S1 (development)
S2 (held-out)
S3 (WHAR Arena models)
RecurrenceNet
S4D
HysteresisNet
QuantileNet
Mamba
No memory
RN + stats
RN + laws
Laws (no mem.)
Stats (no mem.)
17 WHAR models
UTD-MHAD
UCI-HAR
UniMiB-SHAR
ActRecTut
Opportunity
SAD
HAPT
DSADS
WISDM
Daphnet
PAMAP2
UP-Fall
UMAFall
USC-HAD
MotionSense
WEAR
UCA-EHAR
w-HAR
cell: splits × seeds; hatched: excluded by the pre-registered inclusion rule
(c)
test group
training and validation
split
UCI-HAR subject (of 30)